

# Problem Set 21

**Problem 17.2.1:** With  $H^0 = \frac{1}{2m}P^2 + \frac{1}{2}m\omega^2 X^2$  and  $H^1 = \lambda X^4$ , and using that  $X = \sqrt{\hbar/(2m\omega)}(a + a^\dagger)$ , we have:

(1):

$$\begin{aligned}
 E_n^1 &= \langle n | H^1 | n \rangle = \frac{\lambda \hbar^2}{4m^2 \omega^2} \langle n | (a + a^\dagger)^4 | n \rangle \\
 &= \frac{\lambda \hbar^2}{4m^2 \omega^2} \langle n | (a^2 a^{\dagger 2} + a a^\dagger a a^\dagger + a^\dagger a^2 a^\dagger + a a^{\dagger 2} a + a^\dagger a a^\dagger a + a^{\dagger 2} a^2) | n \rangle \\
 &= \frac{\lambda \hbar^2}{4m^2 \omega^2} \langle n | (6a^2 a^{\dagger 2} - 12a a^\dagger + 3) | n \rangle \\
 &= \frac{\lambda \hbar^2}{4m^2 \omega^2} \{6\langle n+2 | (n+1)(n+2) | n+2 \rangle - 12\langle n+1 | (n+1) | n+1 \rangle + 3\langle n | n \rangle\} \\
 &= \frac{\lambda \hbar^2}{4m^2 \omega^2} \{6(n+1)(n+2) - 12(n+1) + 3\} = \frac{3\hbar^2 \lambda}{4m^2 \omega^2} (2n^2 + 2n + 1),
 \end{aligned}$$

where in the first line we dropped the zero superscripts from the unperturbed eigenstates; in the second line we kept only terms with equal numbers of  $a$ 's and  $a^\dagger$ 's; in the third line we used  $[a, a^\dagger] = 1$ ; and in the fourth line we used  $a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle$ .

(2): For any finite value of  $\lambda$ , as  $n$  gets large

$$\frac{E_n^1}{\Delta E} \sim \frac{\hbar^2 \lambda n^2 / (m^2 \omega^2)}{\hbar \omega} \sim n^2 \frac{\hbar \lambda}{m^2 \omega^3} \gg 1.$$

Physically, at large  $x$ , no matter how small  $\lambda$  is,

$$V(x) = \frac{m\omega^2}{2}x^2 + \lambda x^4 \sim \lambda x^4$$

for  $\lambda \neq 0$ .

**Problem 17.2.2:**  $H = -\vec{\mu} \cdot \vec{B} = -\gamma \vec{S} \cdot \vec{B} = H^0 + H^1$  with  $H^0 = -\gamma B_0 S_z$  and  $H^1 = -\gamma B S_x$ . The  $H^0$  eigenvalues are  $E_\pm^0 = \mp \gamma B_0 \hbar / 2$  with eigenstates  $|\pm\rangle^0$ , the  $S_z$  eigenstates. Then

$$E_\pm^1 = \langle \pm |^0 H^1 | \pm \rangle^0 = -\gamma B \langle \pm |^0 S_x | \pm \rangle^0 = -\frac{\gamma B}{2} \langle \pm |^0 (S_+ + S_-) | \pm \rangle^0 = 0,$$

and

$$\begin{aligned}
 E_\pm^2 &= \sum'_m \frac{|\langle \pm |^0 H^1 | m \rangle^0|^2}{E_\pm^0 - E_m^0} = \frac{|\langle \pm |^0 H^1 | \mp \rangle^0|^2}{E_\pm^0 - E_\mp^0} = \frac{\gamma^2 B^2 |\langle \pm |^0 (S_+ + S_-) | \mp \rangle^0|^2}{4 \mp \gamma B_0 \hbar} \\
 &= \mp \frac{\gamma B^2}{4 B_0 \hbar} |\langle \pm |^0 S_\pm | \mp \rangle^0|^2 = \mp \frac{\gamma B^2}{4 B_0 \hbar} |\langle \pm |^0 \hbar \sqrt{(\frac{1}{2} \mp (\mp \frac{1}{2}))(\frac{3}{2} \pm (\mp \frac{1}{2}))} | \mp \rangle^0|^2 = \mp \frac{\gamma \hbar B^2}{4 B_0},
 \end{aligned}$$

and

$$|\pm\rangle' = \sum_m \frac{|m\rangle^0 \langle m|^0 H^1 |\pm\rangle^0}{E_\pm^0 - E_m^0} = \frac{|\mp\rangle^0 \langle \mp|^0 H^1 |\pm\rangle^0}{E_\pm^0 - E_\mp^0} = -\frac{\gamma B}{2} \frac{|\mp\rangle^0 \langle \mp|^0 S_\mp |\pm\rangle^0}{\mp \gamma B_0 \hbar} = \pm \frac{B}{2B_0} |\mp\rangle^0,$$

where we used that  $\langle \mp|^0 S_\mp |\pm\rangle^0 = \hbar$ , as computed in the previous equation. Therefore

$$E_\pm = E_\pm^0 + E_\pm^1 + E_\pm^2 + \mathcal{O}(B^3) = \mp \frac{\gamma B_0 \hbar}{2} \mp \frac{\gamma B^2 \hbar}{4B_0} = \mp \frac{\gamma B_0 \hbar}{2} \left( 1 + \frac{B^2}{2B_0^2} \right),$$

and

$$|\pm\rangle = |\pm\rangle^0 + |\pm\rangle^1 + \mathcal{O}(B^2) = |\pm\rangle^0 \pm \frac{B}{2B_0} |\mp\rangle^0.$$

Now compare to the exact answer.  $H = -\gamma(B_0 S_z + B S_x) = -\gamma \sqrt{B_0^2 + B^2} \hat{n} \cdot \vec{S}$ , where  $\hat{n} = (B_0 \hat{k} + B \hat{i}) / \sqrt{B_0^2 + B^2}$ . The eigenvalues of  $\hat{n} \cdot \vec{S}$  are  $\pm \hbar/2$  with eigenvectors  $|\pm\rangle$  with

$$\begin{aligned} |+\rangle &= \cos \frac{\theta}{2} e^{-i\phi/2} |+\rangle^0 + \sin \frac{\theta}{2} e^{i\phi/2} |-\rangle^0, \\ |-\rangle &= -\sin \frac{\theta}{2} e^{-i\phi/2} |+\rangle^0 + \cos \frac{\theta}{2} e^{i\phi/2} |-\rangle^0, \end{aligned} \quad (1)$$

where  $|\pm\rangle^0$  are the  $S_z$  eigenvectors. Eqn. (1) can be rewritten as

$$|\pm\rangle = \cos \frac{\theta}{2} e^{\mp i\phi/2} |\pm\rangle^0 \pm \sin \frac{\theta}{2} e^{\pm i\phi/2} |\mp\rangle^0,$$

and

$$\begin{aligned} \cos \frac{\theta}{2} &= \sqrt{\frac{1 + \cos \theta}{2}} = \sqrt{\frac{1 + n_z}{2}} = \sqrt{\frac{\sqrt{B_0^2 + B^2} + B_0}{2\sqrt{B_0^2 + B^2}}} = 1 + \mathcal{O}(B^2), \\ \sin \frac{\theta}{2} &= \sqrt{\frac{1 - \cos \theta}{2}} = \sqrt{\frac{1 - n_z}{2}} = \sqrt{\frac{\sqrt{B_0^2 + B^2} - B_0}{2\sqrt{B_0^2 + B^2}}} = \frac{B}{2B_0} + \mathcal{O}(B^3), \end{aligned}$$

and  $\phi = 0$ , since  $\hat{n}$  is in the  $x$ - $z$  plane. This implies

$$|\pm\rangle = |\pm\rangle^0 \pm \frac{B}{2B_0} |\mp\rangle^0 + \mathcal{O}(B^2),$$

in agreement with the perturbation theory result. Similarly, the exact eigenvalues are

$$E_\pm = \mp \frac{\gamma \hbar}{2} \sqrt{B_0^2 + B^2} = \mp \frac{\gamma \hbar B_0}{2} \left( 1 + \frac{B^2}{B_0^2} \right)^{1/2} = \mp \frac{\gamma \hbar B_0}{2} \left( 1 + \frac{B^2}{2B_0^2} + \mathcal{O}(B^4) \right),$$

again in agreement with the perturbation theory result.

**Problem 17.3.2:**  $H = AS_z^2 + B(S_x^2 - S_y^2)$  on the spin-1 Hilbert space. In the  $S_z$  basis,  $|m\rangle$  ( $m = \pm 1, 0$ ), we have

$$S_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad S_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, \quad S_y = \frac{i\hbar}{\sqrt{2}} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$$

(see p. 328 of the text). This implies

$$H = \hbar^2 \begin{pmatrix} A & 0 & B \\ 0 & 0 & 0 \\ B & 0 & A \end{pmatrix}. \quad (2)$$

Clearly

$$|m\rangle = |0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

is the eigenvector with  $E = 0$ . So we only need to look at the  $|m\rangle = |\pm 1\rangle$  subspace where  $H = H^0 + H^1$  with

$$H^0 = \hbar^2 A \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad H^1 = \hbar^2 B \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \text{on the } |m\rangle = \{|\pm 1\rangle\} \text{ subspace.}$$

$H^0$  is degenerate:  $H^0|m\rangle = \hbar^2 A|m\rangle$  for  $m = \pm 1$ . The basis stable under  $H^1$  is the one which diagonalizes  $H^1$ . Since

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = - \begin{pmatrix} 1 \\ -1 \end{pmatrix},$$

the eigenvectors of  $H^1$  are

$$|\bar{1}\rangle \equiv \frac{1}{\sqrt{2}} (|1\rangle + |-1\rangle) \quad \text{and} \quad |\overline{-1}\rangle \equiv \frac{1}{\sqrt{2}} (|1\rangle - |-1\rangle).$$

To order  $\mathcal{O}(B)$ , the energy shifts of  $|\bar{1}\rangle$  and  $|\overline{-1}\rangle$  are

$$\begin{aligned} E_{\bar{1}}^1 &= \langle \bar{1} | H^1 | \bar{1} \rangle = \hbar^2 B \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \frac{1}{\sqrt{2}} = \hbar^2 B, \\ E_{\overline{-1}}^1 &= \langle \overline{-1} | H^1 | \overline{-1} \rangle = \hbar^2 B \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \frac{1}{\sqrt{2}} = -\hbar^2 B. \end{aligned}$$

So, the eigenvalues, to  $\mathcal{O}(B)$ , are

$$E_{\bar{1}} = \hbar^2(A + B), \quad E_{\overline{-1}} = \hbar^2(A - B), \quad E_0 = 0.$$

Compare this to the exact eigenvalues of (2) given by  $0 = \det(H - \lambda) = \lambda(\hbar^2(B - A) + \lambda)(\hbar^2(B + A) - \lambda)$ , which implies  $\lambda = \{0, \hbar^2(A - B), \hbar^2(A + B)\}$ . So, the  $\mathcal{O}(B)$  results are exact.